Milestone 2: Data Handling & Sketches

Congestion Pricing in New York City: What Changed After the Toll?

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Code: github.com/likhith2366/cyro

Overview of data handling

All data is public and is downloaded by `scripts/fetch_data.py`. Large tables (CRZ entries: 10-minute bins; subway: 168M hourly rows; OD estimates: 120M rows/year) are aggregated *on the server* with SoQL (`$select ... sum() ... $group`, `within_polygon()`) so only small summaries are downloaded. `scripts/process_data.py` then builds one processed table per question in `data/processed/`. Common steps: timestamps are treated as New York local time; hours are labelled peak (weekdays 5 a.m.-9 p.m., weekends 9 a.m.-9 p.m.) or off-peak and weekday/weekend; points (crashes, stations, Citi Bike docks) are assigned to **CRZ** (the MTA geofence polygon), **Adjacent** (within 1.5 km of the zone boundary) or **Rest of NYC**; seasonal effects are handled by comparing the same calendar months across years; partial months are dropped.

Datasets added since the proposal: MTA Vehicle Miles Traveled in the CBD (pre-toll baseline for Q2), MTA Subway Origin-Destination Ridership Estimates 2024/2025 (Q6), MTA CBD Geofence (zone outline) and NYC 2020 Neighbourhood Tabulation Areas (maps). **Deferred:** TLC trip records (the CRZ entries feed already has a TLC Taxi/FHV class). **Known gaps:** speeds end Aug 2025; VMT ends Jun 2026; crash data lags ~1 week; Citi Bike is one month per year so far.

Sketches: all sketches are Vega-Lite specifications rendered from the processed data (the course allows sketches made with data-viz tools). They are drafts of layout and encoding; colours, annotation and final interaction will be refined in D3/Vega-Lite for the final project.

Q1. When and where do vehicles enter the zone on a typical weekday, and which entry points are busiest?

Data: MTA CRZ Vehicle Entries (10-minute bins, Jan 5 2025 - Oct 24 2026). **Transformation:** summed entries server-side (SoQL) to date x hour x entry point; kept Mon-Fri; divided by the number of weekdays to get the average weekday entries per hour; summed hours for a daily total, share of all entries, and busiest hour. Each MTA detection group was given an approximate Manhattan-side coordinate for the map.

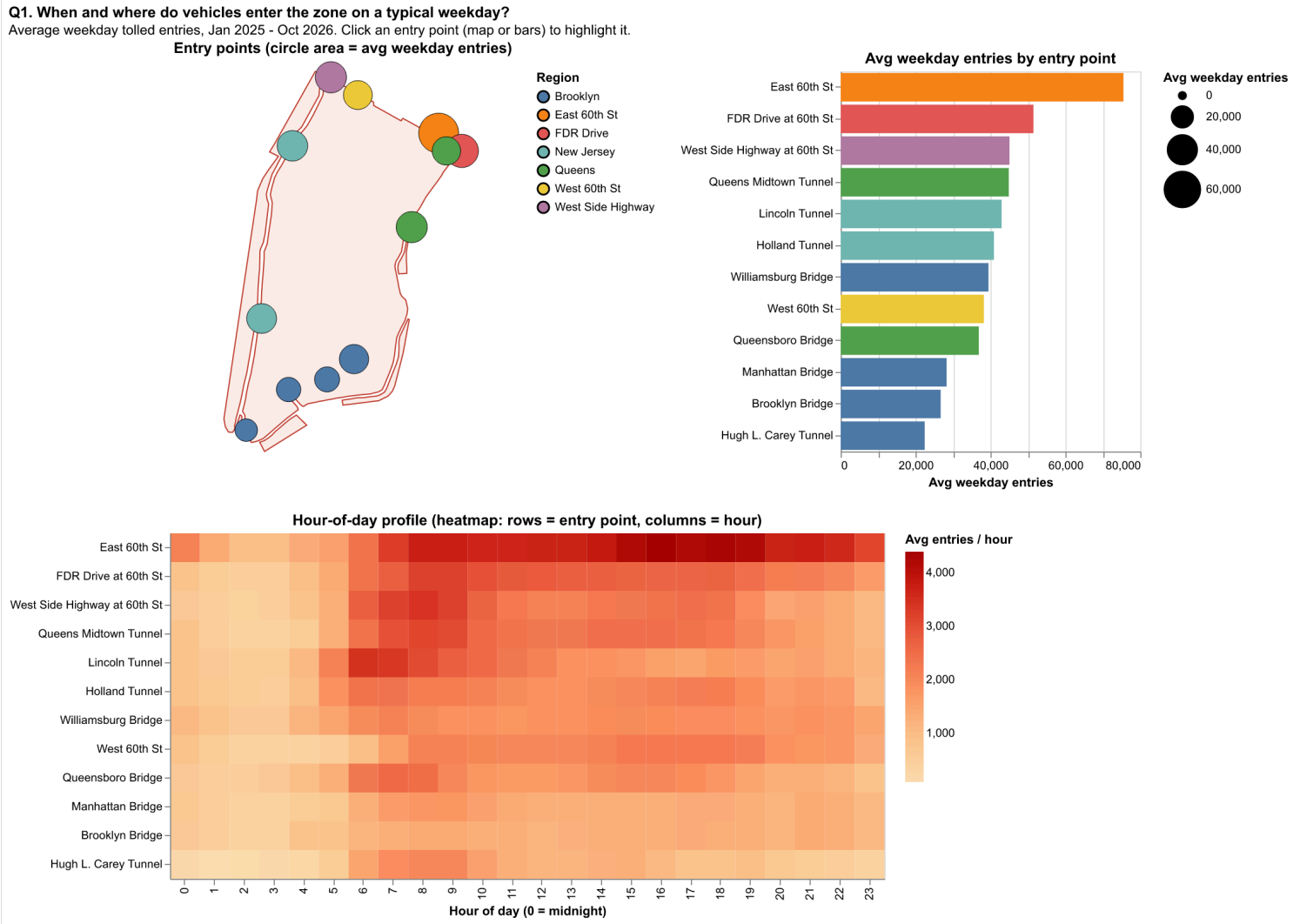
Table. Average weekday tolled entries per entry point. Sample of 12 of 12 rows in `data/processed/q1_entry_point_summary.csv`.

detection_group	detection_region	avg_weekday_crz_entries	share_pct	peak_hour
East 60th St	East 60th St	75,517	15.30	16
FDR Drive at 60th St	FDR Drive	51,462	10.40	9
West Side Highway at 60th St	West Side Highway	45,081	9.10	8
Queens Midtown Tunnel	Queens	44,876	9.10	8
Lincoln Tunnel	New Jersey	42,957	8.70	6
Holland Tunnel	New Jersey	40,914	8.30	6
Williamsburg Bridge	Brooklyn	39,455	8	7
West 60th St	West 60th St	38,221	7.80	18
Queensboro Bridge	Queens	36,868	7.50	7
Manhattan Bridge	Brooklyn	28,285	5.70	9
Brooklyn Bridge	Brooklyn	26,679	5.40	9
Hugh L. Carey Tunnel	Brooklyn	22,425	4.60	8

Table. Hourly profile (Lincoln Tunnel, 5-10 a.m. shown). Sample of 6 of 288 rows in `data/processed/q1_weekday_hourly_by_entry_point.csv`.

detection_group	hour_of_day	avg_crz_entries	avg_excluded_entries
Lincoln Tunnel	5	2,146.40	0
Lincoln Tunnel	6	3,452.90	0
Lincoln Tunnel	7	3,335.10	0
Lincoln Tunnel	8	3,037	0
Lincoln Tunnel	9	2,748.70	0
Lincoln Tunnel	10	2,533	0

What we see so far: East 60th St (the avenues crossing 60th St from the Upper East Side) is the single busiest entry (~75k/weekday, 15% of entries) and peaks in the afternoon (4 p.m.). The New Jersey tunnels peak very early (6 a.m.), Brooklyn and Queens crossings at 7-9 a.m.



Sketch. Linked map + bar chart (click an entry point to highlight it) above an entry-point x hour heatmap. Circle area and bar length = average weekday entries; colour = origin region; heatmap colour = entries per hour. Interactive version: `sketches/q1_entries_when_where.vl.json` / `sketches/index.html`.

Q2. How has the number of entries changed since the toll started, by vehicle class and peak vs. off-peak?

> **Change from proposal:** The entries dataset begins on the day the toll started, so it has *no pre-toll baseline*. We split the question: (a) trends *since* the toll, comparing the same months of 2026 vs 2025 by vehicle class and time period; (b) before vs after, using a dataset we added: **MTA Vehicle Miles Traveled in the CBD (Beginning 2024)**.

Transformation: summed entries to date x vehicle class x time period (MTA's Peak / Overnight label); grouped to month and divided by days in the month (average daily entries); dropped incomplete months (January 2025 starts on the 5th, October 2026 is partial); compared the months present in both years (Feb-Sep). VMT is used as published (monthly average daily VMT for the zone and for the excluded roadways).

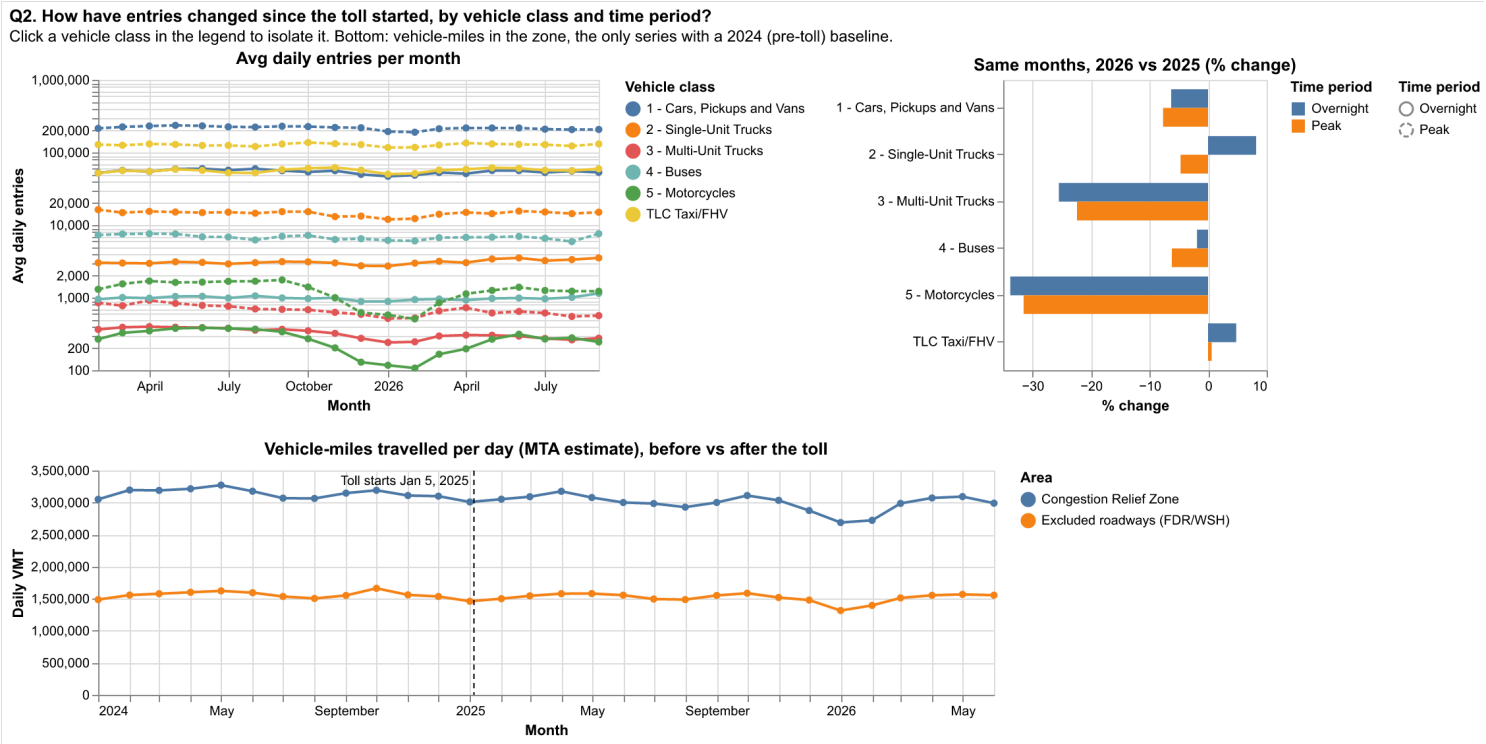
Table. Average daily entries, Feb-Sep 2025 vs Feb-Sep 2026. Sample of 12 of 12 rows in `data/processed/q2_yoy_by_class_period.csv`.

vehicle_class	time_period	avg_daily_2025	avg_daily_2026	pct_change	months_compared
1 - Cars, Pickups and Vans	Overnight	56,880	53,308	-6.30	2-9
1 - Cars, Pickups and Vans	Peak	227,238	209,722	-7.70	2-9
2 - Single-Unit Trucks	Overnight	3,028	3,276	8.20	2-9
2 - Single-Unit Trucks	Peak	15,168	14,461	-4.70	2-9
3 - Multi-Unit Trucks	Overnight	379	282	-25.50	2-9
3 - Multi-Unit Trucks	Peak	788	612	-22.40	2-9
4 - Buses	Overnight	1,007	988	-1.90	2-9
4 - Buses	Peak	7,111	6,670	-6.20	2-9
5 - Motorcycles	Overnight	348	230	-33.80	2-9
5 - Motorcycles	Peak	1,613	1,104	-31.50	2-9
TLC Taxi/FHV	Overnight	55,109	57,728	4.80	2-9
TLC Taxi/FHV	Peak	126,900	127,620	0.60	2-9

Table. Daily VMT inside the zone, Jan-Mar 2024 vs Jan-Mar 2025. Sample of 6 of 60 rows in `data/processed/q2_vmt_monthly.csv`.

date	area	daily_vmt_agps	indexed_average_vmt	post_toll
2024-01-01	Congestion Relief Zone	3,050,435	100	False
2024-02-01	Congestion Relief Zone	3,195,555	102.07	False
2024-03-01	Congestion Relief Zone	3,189,275	104.63	False
2025-01-01	Congestion Relief Zone	3,008,760	90.20	True
2025-02-01	Congestion Relief Zone	3,052,351	92.00	True
2025-03-01	Congestion Relief Zone	3,090,801	92.28	True

What we see so far: peak-period car entries are ~8% lower in 2026 than in the same months of 2025, multi-unit trucks and motorcycles fell 20-30%, while taxis/FHVs are flat or slightly up. VMT inside the zone averaged 3.15M/day in 2024 and 3.03M/day in 2025 (about -4%).



Sketch. Top left: monthly average daily entries (log scale), colour = vehicle class, dash = peak/overnight, legend click isolates a class. Top right: % change 2026 vs 2025 per class and period. Bottom: daily VMT with the toll start marked. Interactive version: `sketches/q2_entries_change.vl.json` / `sketches/index.html`.

Q3. Are drivers avoiding the toll by using the FDR Drive and West Side Highway?

Transformation: four crossings report both tolled entries and *excluded-roadway* entries (vehicles that stay on the untolled FDR / West Side Highway): FDR at 60th, West Side Highway at 60th, Brooklyn Bridge and the Hugh L. Carey Tunnel. For each we computed `excluded share = excluded / (tolled + excluded)` per month and per weekday hour, and labelled hours as peak (5 a.m.-9 p.m.) or overnight.

Table. Monthly excluded-roadway share at each edge crossing (Jan-Feb 2026 shown). Sample of 8 of 88 rows in `data/processed/q3_excluded_share_monthly.csv`.

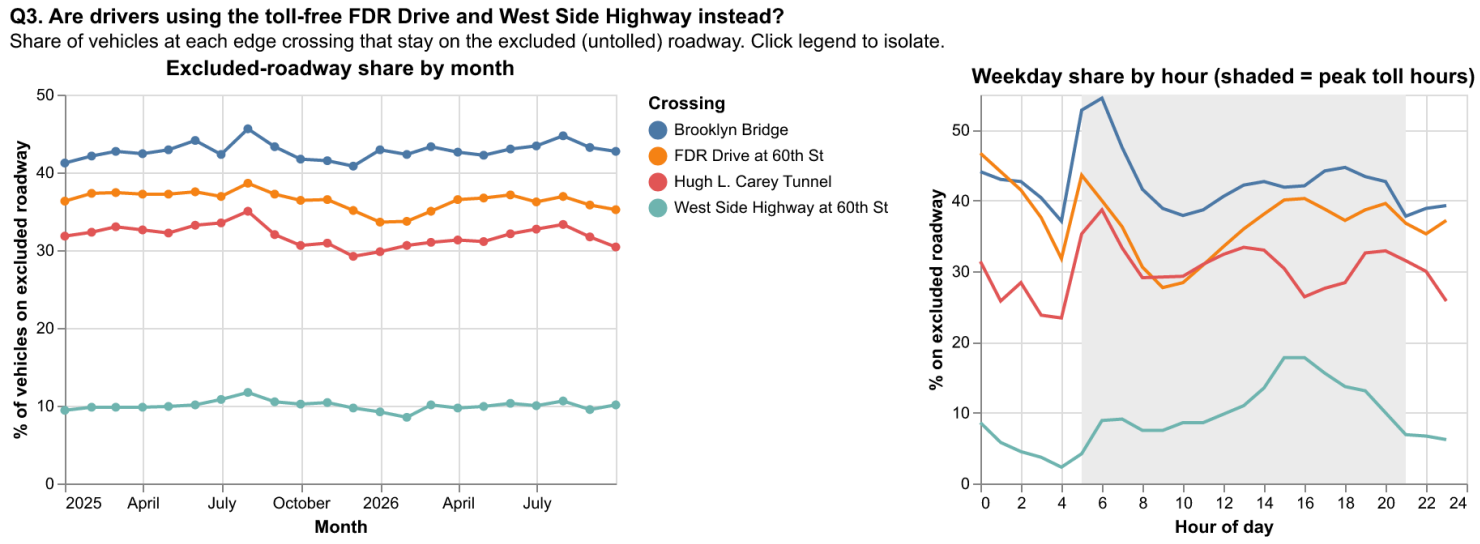
month	detection_group	crz_entries	excluded_roadway_entries	excluded_share_pct
2026-01-01	Brooklyn Bridge	726945	546771	42.90
2026-01-01	FDR Drive at 60th St	1460259	739456	33.60
2026-01-01	Hugh L. Carey Tunnel	588577	250219	29.80
2026-01-01	West Side Highway at 60th St	1158946	117936	9.20
2026-02-01	Brooklyn Bridge	653759	479851	42.30
2026-02-01	FDR Drive at 60th St	1347065	686215	33.70
2026-02-01	Hugh L. Carey Tunnel	542405	238674	30.60
2026-02-01	West Side Highway at 60th St	1098415	101668	8.50

Table. Weekday hourly share, West Side Highway. Sample of 6 of 96 rows in `data/processed/q3_excluded_share_by_hour_weekday.csv`.

detection_group	hour_of_day	toll_period	excluded_share_pct
West Side Highway at 60th St	3	Overnight	3.70
West Side Highway at 60th St	8	Peak	7.50
West Side Highway at 60th St	13	Peak	11
West Side Highway at 60th St	15	Peak	17.80

detection_group	hour_of_day	toll_period	excluded_share_pct
West Side Highway at 60th St	16	Peak	17.80
West Side Highway at 60th St	22	Overnight	6.70

What we see so far: the shares are stable month to month (no growing shift onto the free roads), but at the West Side Highway the excluded share almost doubles in peak hours (11.1% vs 5.9% overnight), which is the kind of time-of-day signal that toll avoidance would produce. We will treat this as a hypothesis, not proof, because the hourly mix of trip purposes also changes.



Sketch. Left: monthly excluded-roadway share per crossing (legend click isolates one). Right: weekday share by hour with the peak toll window shaded. Interactive version: `sketches/q3_toll_avoidance.vl.json` / `sketches/index.html`.

Q4. Has traffic inside the zone become faster than in 2019-2024, and than the rest of the city?

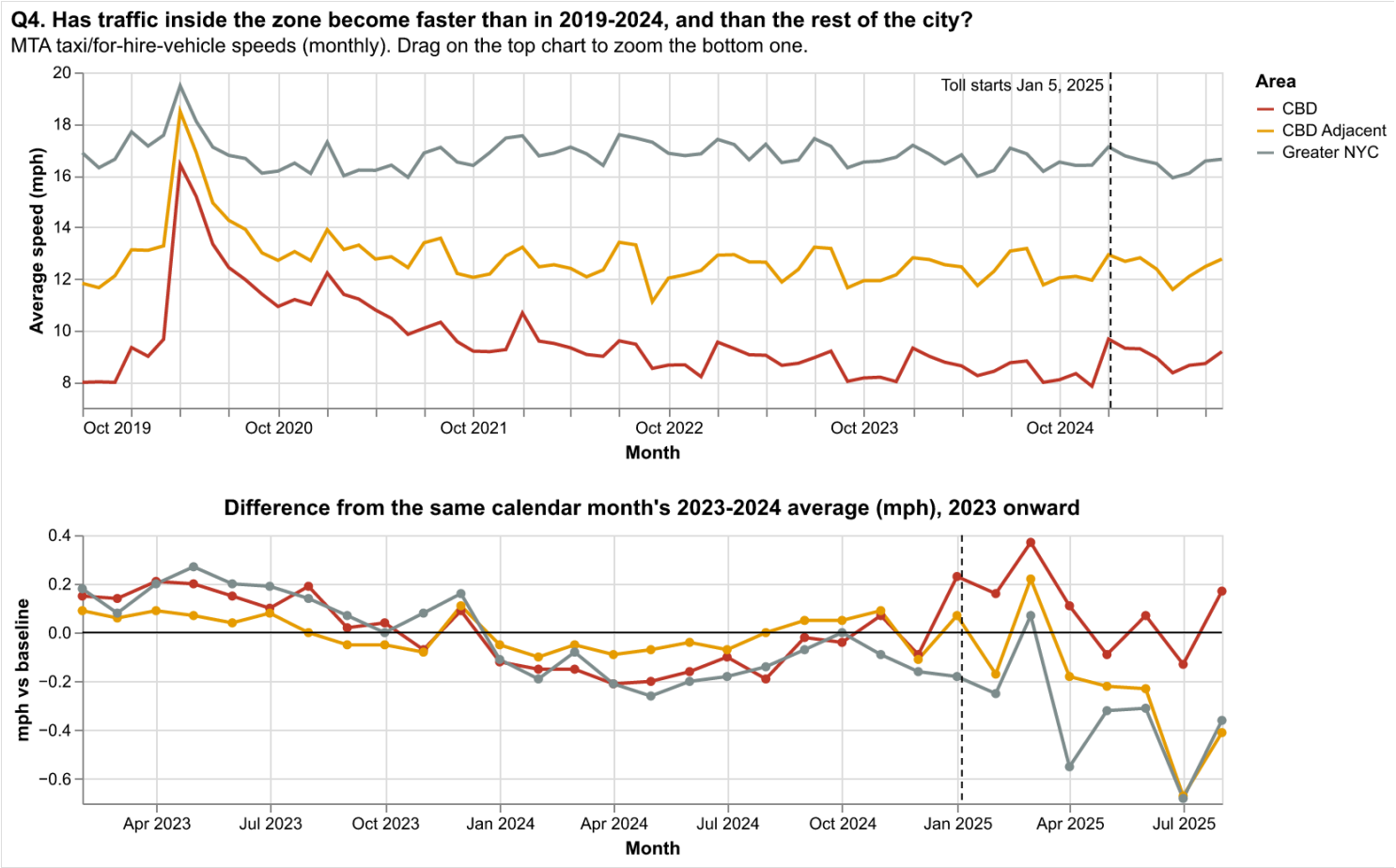
> **Change from proposal:** The MTA speed dataset is **monthly taxi/FHV speeds** for three areas (CBD, CBD Adjacent, Greater NYC), not hourly speeds for all vehicles, and it currently ends in **August 2025**. Because 2020-2022 speeds are inflated by the pandemic, we compare against the same calendar month in **2023-2024** rather than 2019-2024; the full 2019-2025 series is still shown for context.

Transformation: for each area and calendar month we averaged 2023 and 2024 speeds as a baseline and computed the difference (mph and %) for every month.

Table. Taxi/FHV speeds, Jan-Mar 2025, vs the 2023-24 same-month baseline. Sample of 9 of 213 rows in `data/processed/q4_speeds_monthly.csv`.

month	zone	zonal_speed	baseline_2023_24_mph	diff_vs_baseline_mph	pct_vs_baseline
2025-01-01	CBD	9.66	9.43	0.23	2.40
2025-02-01	CBD	9.30	9.14	0.16	1.80
2025-03-01	CBD	9.28	8.91	0.37	4.20
2025-01-01	CBD Adjacent	12.93	12.86	0.07	0.50
2025-02-01	CBD Adjacent	12.67	12.84	-0.17	-1.30
2025-03-01	CBD Adjacent	12.81	12.60	0.22	1.70
2025-01-01	Greater NYC	17.11	17.28	-0.18	-1
2025-02-01	Greater NYC	16.76	17.02	-0.25	-1.50
2025-03-01	Greater NYC	16.60	16.53	0.07	0.40

What we see so far: CBD speeds were +3.1% higher in Jan-Aug 2025 than Jan-Aug 2024, while the adjacent area and the rest of the city were about 1% slower; the CBD is the only area consistently above its baseline after January 2025.



Sketch. Top: monthly speed by area (toll start marked; drag to brush a time range, legend click isolates an area). Bottom: difference from the 2023-24 same-month baseline, zoomed to the brushed range. Interactive version: `sketches/q4_speeds.vl.json` / `sketches/index.html`.

Q5 + Q7. Did crashes and injuries fall inside the zone, and did they rise just outside it?

> **Change from proposal:** Q5 and Q7 use the same crash data with the same three-area split, so we merged them into one view. Q5's traffic part is answered by Q3/Q4 (excluded roadways and the CBD Adjacent speed series).

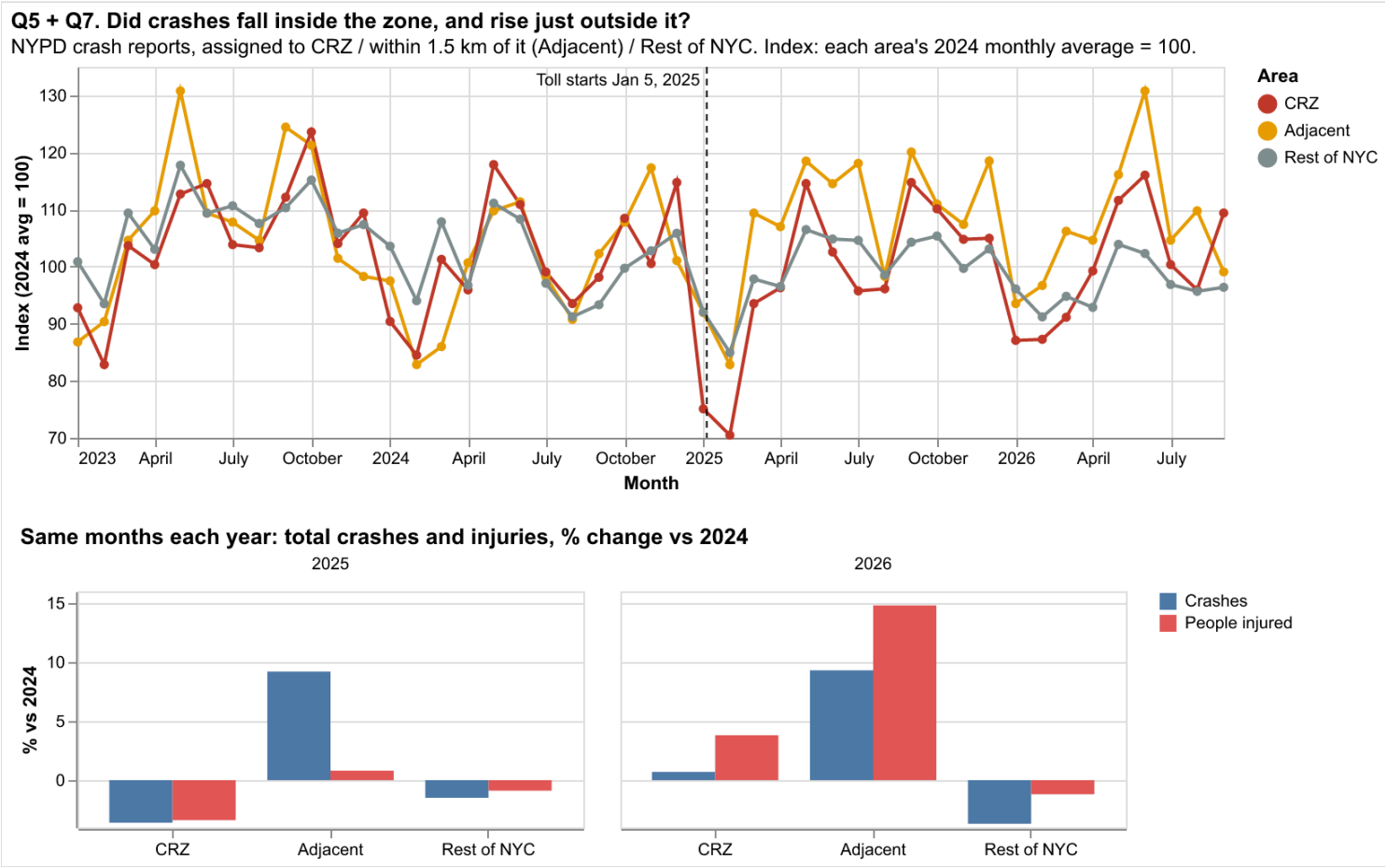
Transformation: downloaded all NYPD crash records since Jan 2023 (342k); dropped 20k with missing or out-of-city coordinates; assigned each crash to **CRZ** (inside the MTA geofence polygon), **Adjacent** (outside but within 1.5 km of the zone boundary: Upper East/West Side to roughly 85th St, Long Island City, Downtown Brooklyn) or **Rest of NYC** by point-in-polygon; aggregated to month x area (crashes, injured, killed, pedestrians, cyclists); compared the same months (Jan-Sep) each year.

Table. Crashes and injuries, Jan-Sep of each year. Sample of 12 of 12 rows in `data/processed/q5_q7_collisions_yoy_by_area.csv`.

area	year	crashes	persons_injured	pedestrians_injured	cyclist_injured	crashes_pct_vs_2024	persons_injured_pct_vs_2024
Adjacent	2023	2444	1146	180	187	10.10	2.60
Adjacent	2024	2219	1117	176	177	0	0
Adjacent	2025	2424	1126	170	187	9.20	0.80
Adjacent	2026	2426	1282	212	190	9.30	14.80
CRZ	2023	5021	2376	628	615	3.90	1.60
CRZ	2024	4833	2338	688	636	0	0
CRZ	2025	4657	2258	677	619	-3.60	-3.40
CRZ	2026	4868	2426	731	642	0.70	3.80
Rest of NYC	2023	58838	33154	5310	2873	6.60	0.60

area	year	crashes	persons_injured	pedestrians_injured	cyclist_injured	crashes_pct_vs_2024	persons_injured_pct_vs_2024
Rest of NYC	2024	55217	32952	5360	2700	0	0
Rest of NYC	2025	54415	32643	5364	2602	-1.50	-0.90
Rest of NYC	2026	53190	32570	5305	2590	-3.70	-1.20

What we see so far: in 2025 crashes fell 3.6% inside the zone and rose 9.2% in the adjacent ring (vs -1.5% citywide). In 2026 the CRZ is back near its 2024 level while the adjacent ring stays about 9% higher. The monthly index is noisy (e.g. a snowy February 2025), so the final view will show monthly values, not only totals.



Sketch. Top: monthly crashes indexed to each area's 2024 average (a dropdown switches the measure to injuries, pedestrians or cyclists). Bottom: % change vs 2024 for the same months. Interactive version: `sketches/q5_q7_crashes.vl.json` / `sketches/index.html`.

Q6. Did more people take the subway or Citi Bike into the zone, and from which neighbourhoods?

> **Change from proposal:** Station tap counts cannot tell where a rider is going, so we added the **MTA Subway Origin-Destination Ridership Estimate** (2024 and 2025) to answer the "from which neighbourhoods" part. Citi Bike monthly files are 0.4-1 GB each, so for this milestone we processed one matched pair of months (February 2024 vs February 2025).

Transformation: (1) subway hourly ridership summed to station x month and classified by station location (CRZ / Adjacent / Rest); (2) OD estimates filtered server-side with `within_polygon()` to weekday trips that *end* inside the zone and start outside it, for October 2024 and October 2025; averaged to an average weekday; origin stations joined to 2020 Neighbourhood Tabulation Areas; (3) Citi Bike: removed trips without coordinates, shorter than 1 minute or longer than 3 hours (~0.3%), kept trips ending in the zone, and grouped by start neighbourhood.

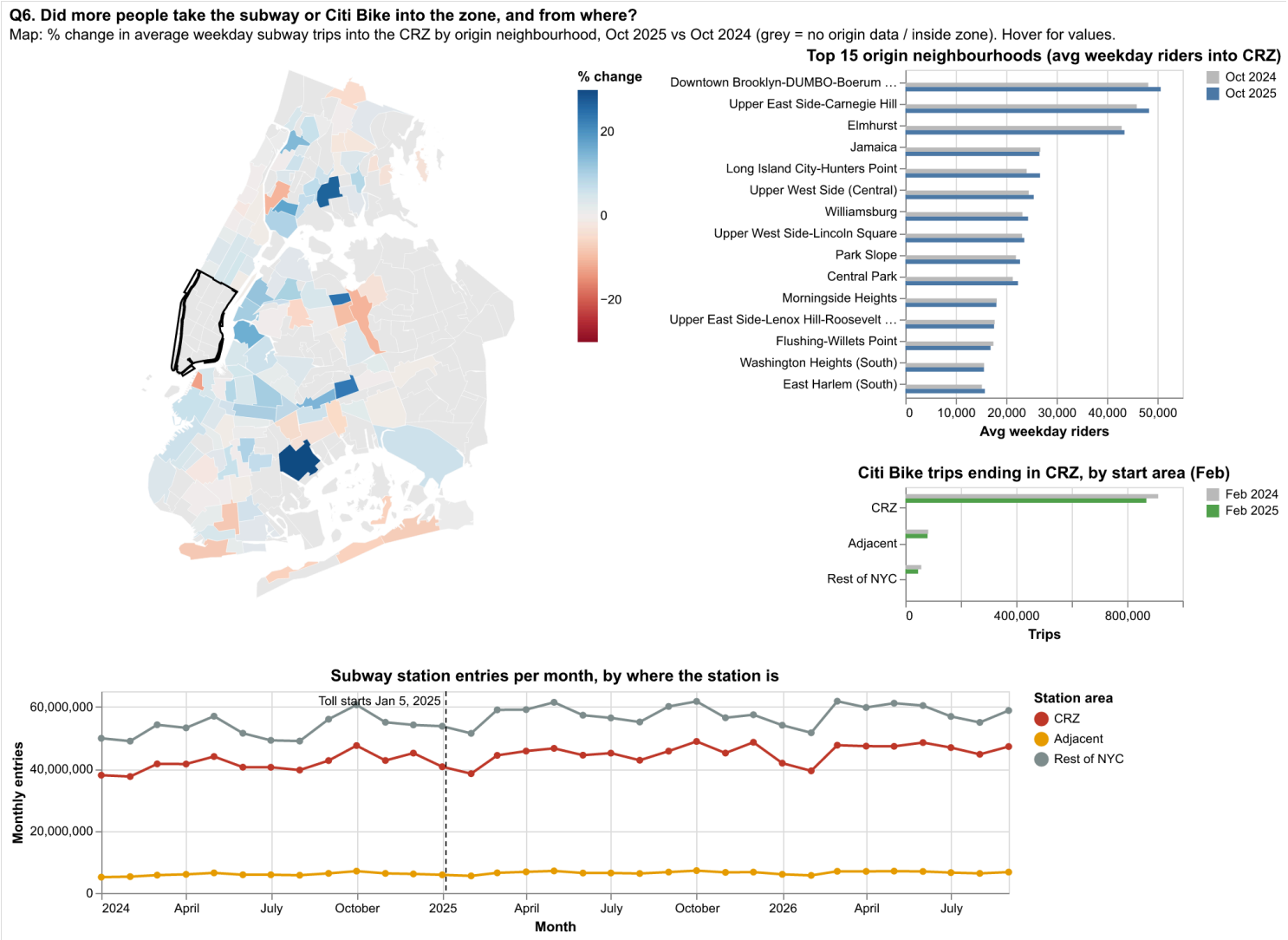
Table. Average weekday subway trips into the zone by origin neighbourhood. Sample of 8 of 117 rows in `data/processed/q6_subway_od_into_crz_by_neighborhood.csv`.

nta	borough	avg_weekday_riders_oct2024	avg_weekday_riders_oct2025	change	pct_change
Downtown Brooklyn-DUMBO-Boerum Hill	Brooklyn	48,188.30	50,664.90	2,476.60	5.10
Upper East Side-Carnegie Hill	Manhattan	45,919.20	48,331.10	2,411.90	5.30
Elmhurst	Queens	42,909.70	43,480.90	571.20	1.30
Long Island City-Hunters Point	Queens	24,078.10	26,725.80	2,647.70	11
Jamaica	Queens	26,795.10	26,596	-199.20	-0.70
Upper West Side (Central)	Manhattan	24,479.20	25,473.90	994.80	4.10
Williamsburg	Brooklyn	23,226	24,354.10	1,128	4.90
Upper West Side-Lincoln Square	Manhattan	23,146.80	23,588.60	441.70	1.90

Table. Citi Bike trips ending in the zone, by start area. Sample of 3 of 3 rows in `data/processed/q6_citibike_into_crz_by_start_area.csv`.

start_area	trips_feb2024	trips_feb2025	pct_change
Adjacent	82540	80013	-3.10
CRZ	912155	869521	-4.70
Rest of NYC	57408	46039	-19.80

What we see so far: average weekday subway trips into the zone rose ~3.4% (985k to 1.02M) between October 2024 and October 2025, with the largest absolute gains from Brooklyn. However, station entries rose 7-9% in *every* area in 2025, so much of this is citywide ridership recovery. Citi Bike trips into the zone were lower in February 2025 than February 2024; one winter month is not enough, so we will add more months.



Sketch. Left: choropleth of % change in weekday subway trips into the zone by origin neighbourhood (zone outlined; hover for values). Right: top origin neighbourhoods and Citi Bike trips by start area. Bottom: monthly subway entries by station area. Interactive version: `sketches/q6_transit_into_zone.vl.json` / `sketches/index.html`.

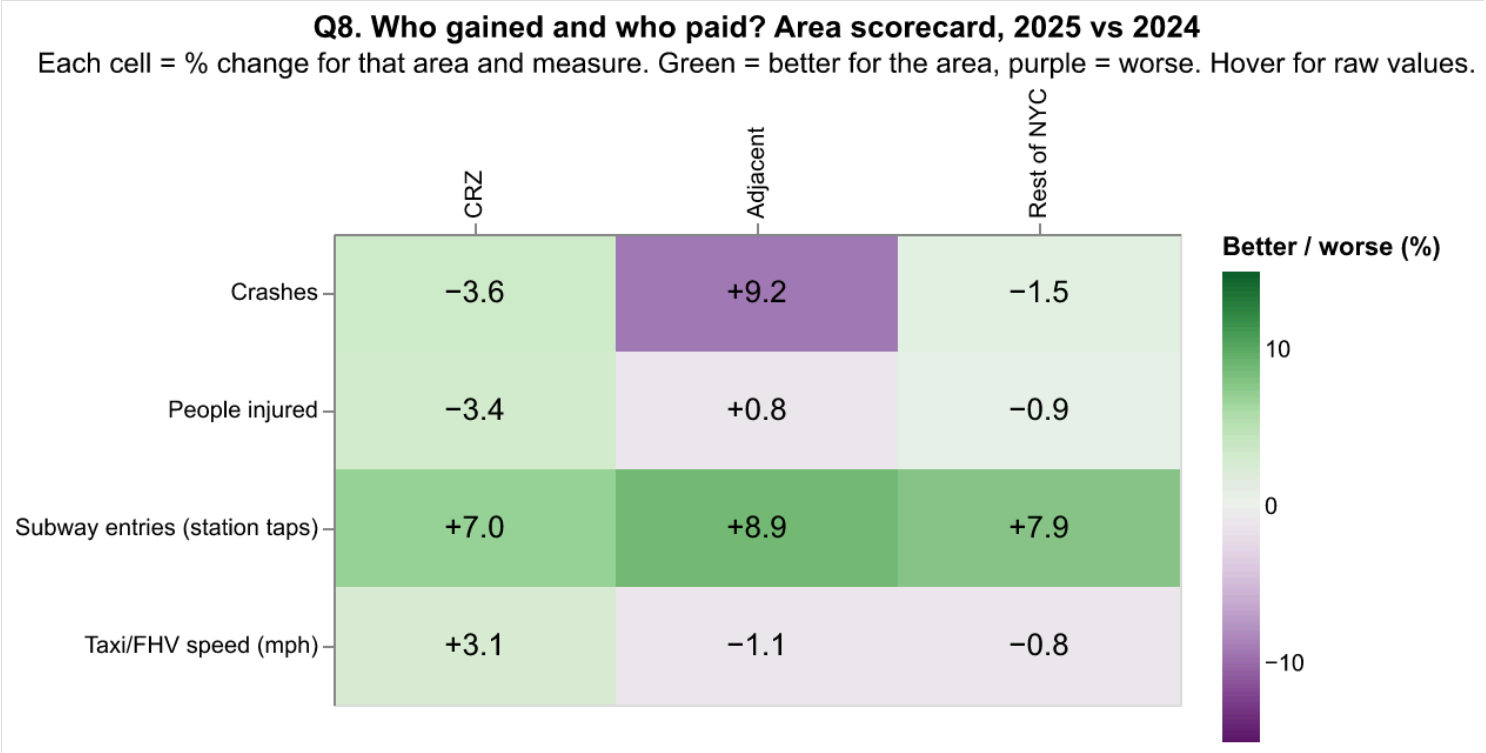
Q8. Putting it together: which areas gained, and which carried the costs?

Transformation: joined the per-area results from Q4, Q5/Q7 and Q6 into one long table (area x measure, 2024 vs 2025 over the months each dataset covers) and computed % change; for display, the sign is flipped for crashes and injuries so that green always means "better for that area".

Table. Area scorecard. Sample of 12 of 12 rows in `data/processed/q8_area_scorecard.csv`.

area	metric	window	value_2024	value_2025	pct_change
CRZ	Taxi/FHV speed (mph)	months 1-8	8.74	9.01	3.10
Adjacent	Taxi/FHV speed (mph)	months 1-8	12.60	12.46	-1.10
Rest of NYC	Taxi/FHV speed (mph)	months 1-8	16.66	16.52	-0.80
CRZ	Crashes	Jan-Sep	4,833	4,657	-3.60
CRZ	People injured	Jan-Sep	2,338	2,258	-3.40
Adjacent	Crashes	Jan-Sep	2,219	2,424	9.20
Adjacent	People injured	Jan-Sep	1,117	1,126	0.80
Rest of NYC	Crashes	Jan-Sep	55,217	54,415	-1.50
Rest of NYC	People injured	Jan-Sep	32,952	32,643	-0.90

area	metric	window	value_2024	value_2025	pct_change
CRZ	Subway entries (station taps)	Jan-Dec	501,163,149	536,213,455	7
Adjacent	Subway entries (station taps)	Jan-Dec	71,807,923	78,169,367	8.90
Rest of NYC	Subway entries (station taps)	Jan-Dec	638,541,831	689,252,002	7.90



Sketch. Area x measure heatmap; colour = better (green) / worse (purple), label = % change, tooltip = raw values. In the final version clicking a cell will open the Q1-Q6 chart behind it, and we plan a neighbourhood-level map version. Interactive version: `sketches/q8_area_scorecard.vl.json` / `sketches/index.html`.

AI-use statement (per course policy): *We used Claude (Anthropic) as a coding assistant to help write the data download / processing scripts and the Vega-Lite specifications for the sketches, and to help draft parts of this report. The domain questions, the choice of data transformations and chart designs, and the interpretation of results were reviewed, checked against the data, and edited by the team. The same statement is in the repository README.*